



AI-assisted Software Development

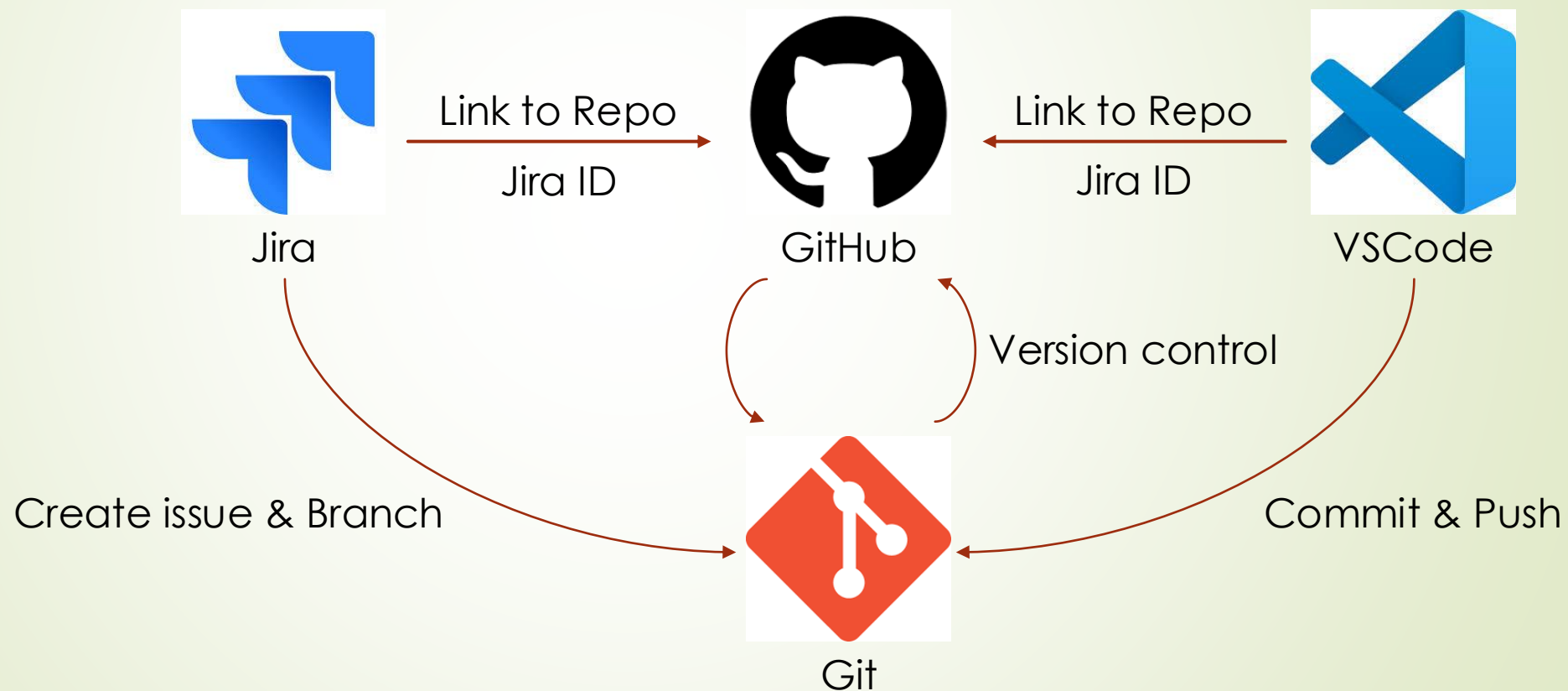
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Technology Involved

- VSCode
 - Integrated Development Environment (Editor + Compiler/Interpreter + Debugging + Git Integration)
- GitHub
 - Cloud-based source code repository and collaboration platform for version control, branch management, pull requests, and code sharing
- Git
 - Distributed version control tool for tracking code changes, creating branches, committing updates, and synchronizing with remote repositories
- Jira
 - Project management and issue tracking tool for planning tasks, managing workflows, and linking development work with GitHub branches and commits

Development Workflow

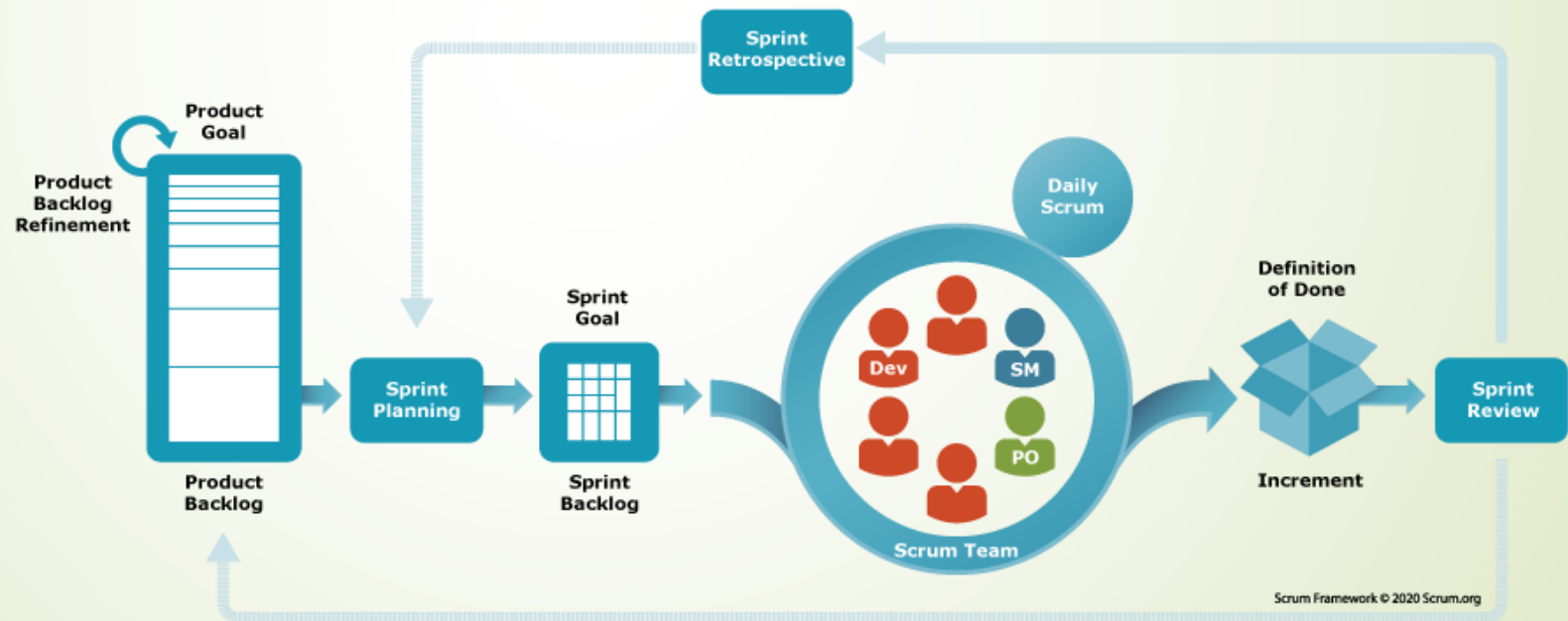


Jira

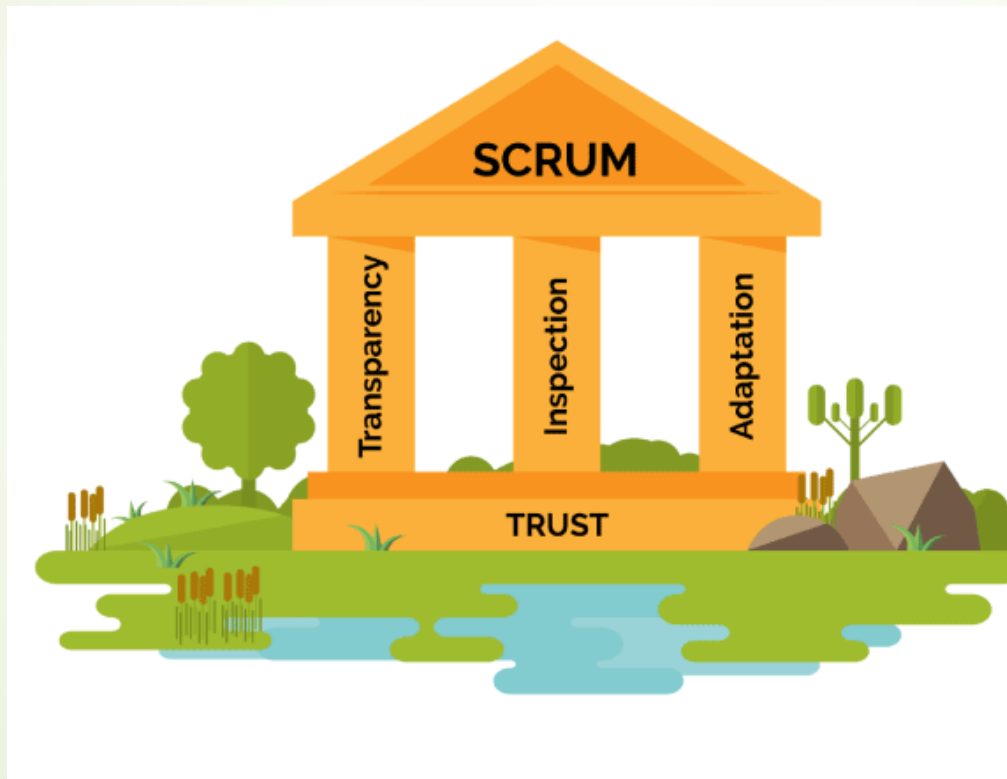
- Spaces: 11402_CS351
- Four projects == four epics
 - Project0
 - ProjectA
 - ProjectB
 - ProjectC

Scrum (scrum.org)

- As an **agile framework**, Scrum provides just enough structure for people and teams to integrate into how they work, while adding the right practices to optimize for their specific needs.



Scrum Values



COURAGE

Scrum Team members have courage to do the right thing and work on tough problems



FOCUS

Everyone focuses on the work of the Sprint and the goals of the Scrum Team



COMMITMENT

People personally commit to achieving the goals of the Scrum Team



RESPECT

Scrum Team members respect each other to be capable, independent people



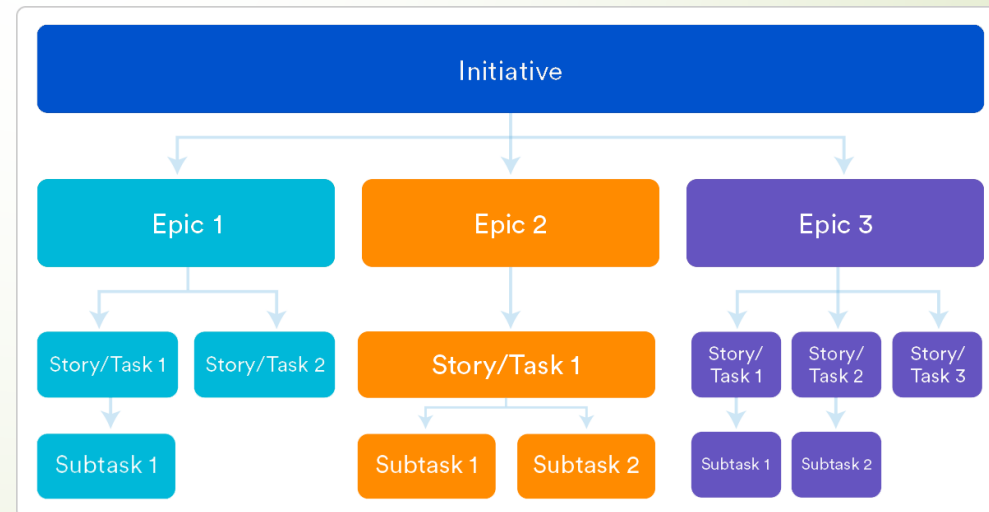
OPENNESS

The Scrum Team and its stakeholders agree to be open about all the work and the challenges with performing the work

Credit: ABN AMRO Bank N.V.

Scrum in Jira

- Epic: A big body of work (e.g., "Implement User Authentication") lasting several sprints.
- Story: A user-centric feature (e.g., "As a user, I want to login with email").
- Task: Specific technical steps (e.g., "Create database schema for users").
- Subtask: Further breaks down a Task or Story into even smaller pieces, like "Design login UI"



Before Development & Coding

- Communication matters first
- Coding is only one piece of software development
- Shared conventions reduce confusion
- Naming and style improve teamwork
- Good code should be readable by humans and AI

Alignment What?

Before development and coding, a team must first align on:

- problem definition
- requirements
- task scope
- naming rules
- coding style
- workflow
- responsibilities
- expected outputs

Without this shared understanding, even technically correct code can become hard to read, hard to review, and hard to maintain.

Why Coding Style & Naming Convention Matter

Coding style and naming convention are not only technical rules. They are a form of **team communication**.

They help developers:

- understand each other's code faster
- reduce ambiguity
- make reviews easier
- collaborate with AI tools more consistently
- maintain the project over time

“Write code in a way that other humans and AI collaborators can understand.”

Coding Style & Naming Convention

- Google C++ Style Guide
 - <https://google.github.io/styleguide/cppguide.html>

Example

Category	Google Style	Standard (STL)	Microsoft / Windows
Classes / Structs	PascalCase (MyClass)	snake_case (my_class)	PascalCase (MyClass)
Functions / Methods	PascalCase (DoSomething)	snake_case (do_something)	PascalCase (DoSomething)
Variables (Local)	snake_case (my_var)	snake_case (my_var)	camelCase (myVar)
Member Variables	snake_case_ (my_var_)	snake_case (my_var)	m_prefix (m_myVar)
Constants	kPascalCase (kDaysInWeek)	snake_case (days_in_week)	SCREAMING_SNAKE (MAX_VAL)
Files	snake_case.cc	snake_case.cpp	PascalCase.cpp
Namespaces	snake_case	snake_case	PascalCase

Sprint

- In Scrum, the team does not simply “start coding.”
- A sprint begins with agreeing on **what should be done, why it matters**, and **what counts as done**.
- During the sprint, the team keeps communicating, tracks progress, solves blockers, reviews the result, and reflects on how to improve the next cycle.

A sprint is a communication-and-delivery loop, not just a coding period.

Sprint = Plan → Align → Build → Review → Improve

- Sprint Planning: decide the goal and tasks
- Daily Scrum: sync progress and blockers
- Development Work: design, coding, testing, integration, documentation
- Sprint Review: show what was completed
- Sprint Retrospective: improve teamwork and process

Take Home Message

In real-world software development, success depends not only on writing code, but also on communicating clearly before coding starts. Requirements, workflow, naming conventions, and coding style help a team build software in a consistent and maintainable way. Coding is only one part of software development; communication is what connects all the parts together.

Project 0: Two Sum

► Problem

► [LeetCode 1](#)

- Given an array of integers `nums` and an integer `target`, return *indices of the two numbers such that they add up to target*.
- You may assume that each input would have **exactly one solution**, and you may not use the same element twice.

► Implementation

► Two Sum by Array

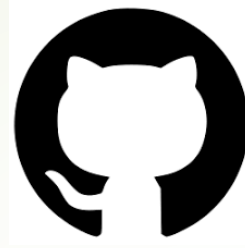
- `vector<int> TwoSumArray(vector<int>& nums, int target)`

► Two Sum by HashTable

- `vector<int> TwoSumHashTable(vector<int>& nums, int target)`



Remote/cloud



local

